

# Growth and Decay

GCSE Maths · Practice Questions

20 questions · 62 marks total

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## Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Write down the single multiplier for a 5% increase.

.....

2 [2 marks]

£500 increases by 10%.

Work out the new amount.

.....

3 [2 marks]

A population of 8000 falls by 25%.

Work out the new population.

.....

4 [2 marks]

£1000 is invested for 2 years at 3% **simple** interest.

Work out the total amount at the end.

.....

5 [3 marks]

£1000 is invested for 2 years at 3% **compound** interest.

(a) Work out the total amount at the end.

(b) How much more is this than with simple interest at the same rate?

.....

6

[3 marks]

A car is worth £12 000. Its value falls by 20% each year.  
Work out its value after 2 years.

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7

[3 marks]

A colony of bacteria doubles in size every hour.  
There are 500 bacteria at the start.  
How many are there after 5 hours?

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8

[3 marks]

£2500 is invested at 4% compound interest per year.  
Work out the value after 3 years, to the nearest penny.

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9

[4 marks]

£5000 is invested for 3 years. Two accounts are available.  
**Account A:** 4% simple interest per year.  
**Account B:** 3.5% compound interest per year.  
Which account gives more money after 3 years, and by how much?

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10

[4 marks]

After a 15% decrease, the value of an item is £510.  
(a) Work out the original value.  
(b) Work out the size of the decrease in pounds.

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Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

**1** [2 marks]

Write down the single multiplier for a 12.5% decrease.

.....

**2** [3 marks]

£8000 is invested at 2% compound interest per year.  
Work out the value after 5 years, to the nearest penny.

.....

**3** [3 marks]

A radioactive substance decays by 8% each day.  
There are 400 g at the start.  
Work out the mass after 10 days, to 1 decimal place.

.....

**4** [3 marks]

An investment of £4000 grows to £4630.50 in 3 years with compound interest.  
Work out the annual interest rate.

.....

**5** [4 marks]

£6000 is invested at 3% compound interest per year.  
Work out the least number of complete years before the investment exceeds £8000.

.....

6

[4 marks]

A machine is worth £40 000 when new.

It loses 25% of its value in the first year and 15% of its value in each year after that.

Work out its value after 4 years.

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7

[4 marks]

The number of users  $N$  of an app after  $t$  months is modelled by

$$N = 500 \times 1.4^t$$

(a) How many users were there at the start?

(b) How many users after 6 months? Give your answer to the nearest whole number.

(c) What does the number 1.4 tell you?

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8

[4 marks]

A substance has a half-life of 12 years, so its mass halves every 12 years.

There are 640 g at the start.

(a) Work out the mass after 48 years.

(b) Write a formula for the mass  $m$  grams after  $t$  years.

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9

[4 marks]

£5000 is invested for one year. Two options are offered.

**Option 1:** 6% interest added once, at the end of the year.

**Option 2:** 0.5% interest added at the end of every month.

Which option gives more money, and by how much?

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An amount £P is invested at 4% compound interest per year.

After 5 years it is worth £9733.22

(a) Work out P.

(b) Work out how many more complete years it takes for the investment to exceed £12 000.

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